

Multiple Choice

1. **Answer:** A

Options: A) 26.6 days; B) 19.0 days; C) 3.80 days; D) 5.7 days; E) 22.8 days

Note: Correct option: A - 26.6 days

2. **Answer:** A

Options: A) 5 dyne/stat-coul; B) 8 dyne/stat-coul; C) 0.5 dyne/stat-coul; D) 100 dyne/stat-coul; E) 10 dyne/stat-coul

Note: Correct option: A - 5 dyne/stat-coul

3. **Answer:** A

Options: A) 0.0612; B) 0.0426; C) 0.0753; D) 0.0905; E) 0.0294

Note: Correct option: A - 0.0612

4. **Answer:** A

Options: A) 4.8; B) 7.1; C) 5.5; D) 1.8; E) 4.2

Note: Correct option: A - 4.8

5. **Answer:** B

Options: A) 5; B) 7; C) 9; D) 12

Note: Correct option: B - 7

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Matches the MMLU-Pro answer key.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It contains precious metals'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'the work done on them'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'pressure of fluid'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '6 volts'.

Long Form Response

11. **Answer:** $2.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 3.5 m. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.